

Review

A Survey on the Applications of Cloud Computing in the Industrial Internet of Things

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Abstract: The convergence of cloud computing and the Industrial Internet of Things (IIoT) has significantly transformed industrial operations, enabling intelligent, scalable, and efficient systems. This survey provides a comprehensive analysis of the role cloud computing plays in IIoT ecosystems, focusing on its architectural frameworks, service models, and application domains. By leveraging centralized, edge, and hybrid cloud architectures, IIoT systems achieve enhanced real-time processing capabilities, streamlined data management, and optimized resource allocation. Moreover, this study delves into integrating artificial intelligence (AI) and machine learning (ML) in cloud platforms to facilitate predictive analytics, anomaly detection, and operational intelligence in IIoT environments. Security challenges, including secure device-to-cloud communication and privacy concerns, are addressed with innovative solutions like blockchain and AI-powered intrusion detection systems. Future trends, such as adopting 5G, serverless computing, and AI-driven adaptive services, are also discussed, offering a forward-looking perspective on this rapidly evolving domain. Finally, this survey contributes to a well-rounded understanding of cloud computing's multifaceted aspects and highlights its pivotal role in driving the next generation of industrial innovation and operational excellence.

Keywords: cloud computing; Industrial Internet of Things; architectures; predictive maintenance; data security and privacy



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1. Introduction

IIoT has emerged as a cornerstone of ongoing digital transformation in industries, enabling enhanced connectivity, automation, and intelligence in operations. By linking physical systems with advanced digital technologies, IIoT facilitates real-time monitoring, improved efficiency, and innovative business models across the manufacturing, healthcare, energy, and logistics sectors. Central to the success of IIoT is the ability to manage and analyze the massive volumes of data generated by interconnected devices, which has positioned cloud computing as an essential enabler of these systems [1,2].

Cloud computing offers IIoT environments the scalability, flexibility, and computational power needed to process and store the data collected from diverse devices and sensors. Its ability to centralize operations and provide on-demand resources enables industries to optimize workflows and enhance decision-making processes. By reducing the need for extensive on-premises infrastructure, cloud computing also lowers costs and allows organizations to focus on their core operations while benefiting from advanced technological capabilities [3,4].

However, integrating cloud computing with IIoT is not without challenges. The dynamic and distributed nature of IIoT systems requires solutions that can accommodate

real-time data processing and reliable connectivity. At the same time, concerns surrounding data security, privacy, and system reliability remain critical, as industrial operations involve sensitive information and mission-critical tasks. Addressing these challenges requires a comprehensive understanding of how cloud computing can support the unique demands of IIoT systems while ensuring robust and secure operations [5,6].

The interplay between cloud computing and IIoT is shaping the future of industrial systems by unlocking new opportunities for innovation and efficiency. As industries increasingly adopt digital technologies, the role of cloud computing in supporting IIoT will continue to grow. This convergence is expected to redefine industrial processes, offering a pathway toward smarter, more sustainable, and resilient operations that can adapt to the evolving demands of the modern world [7,8].

The difference between the present survey and the surveys in Table 1 lies in its detailed examination of cloud computing architectures specifically tailored for IIoT environments, including centralized, distributed, and hybrid models. Unlike the referenced surveys that focus on general applications or specific technological integrations like blockchain or edge computing, this survey emphasizes the interplay of cloud computing service models (Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS)) and their direct impact on IIoT scalability, latency, and security. It also uniquely explores real-world applications such as predictive maintenance and energy management while addressing advanced trends like AI integration, serverless computing, and 5G-enabled IIoT solutions. Furthermore, the survey delves into innovative security mechanisms like AI-driven intrusion detection and blockchain, offering a forward-looking perspective beyond traditional challenges. In conclusion, this survey achieves the following:

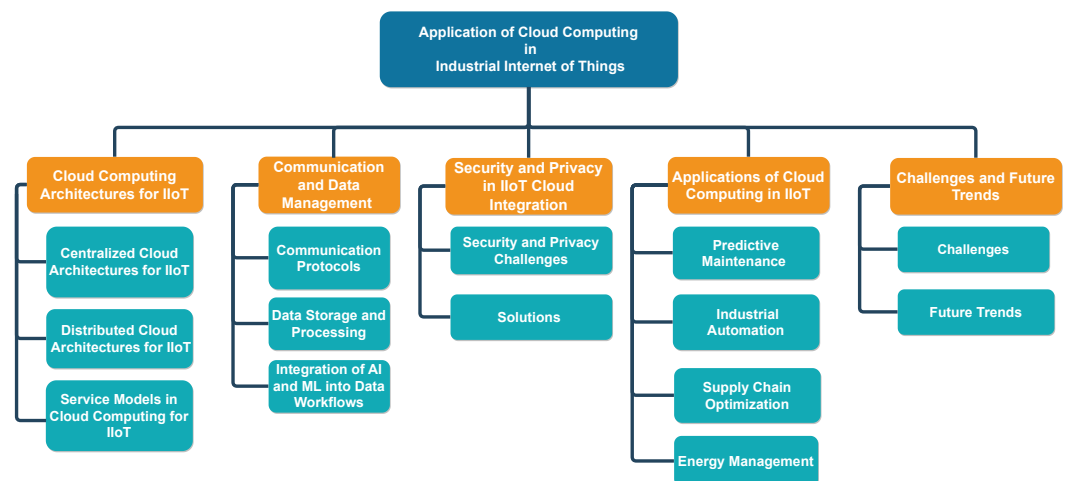
- It provides a concise and focused description of cloud computing architectures, service models, and their applicability to diverse IIoT environments.
- It identifies key challenges in cloud-IIoT integration, including latency, security, and data management, while reviewing innovative mitigation approaches.
- It highlights real-world applications across industries, showcasing the transformative impact of cloud computing in IIoT systems.
- It discusses emerging trends, such as adopting hybrid cloud models and AI-driven cloud services, to guide future research and industrial implementations.

Figure 1 provides an overview of the key topics discussed in this survey, emphasizing the critical aspects of cloud computing integration within IIoT ecosystems. It highlights architectural frameworks, service models, and specific application domains while showcasing security challenges, innovative solutions, and future trends. This visual guide is intended to help readers navigate the comprehensive analysis presented in the survey by linking the identified challenges and opportunities to their respective sections. Each element in the figure aligns with the subsequent discussion in the survey's sections, providing a structured pathway for understanding the transformative role of cloud computing in IIoT.

The following sections are structured as follows. Section 2 discusses cloud computing architectures. Section 3 covers communication protocols and data storage and processing. Section 4 explores security challenges and solutions. Section 5 outlines applications in predictive maintenance, industrial automation, supply chains, and energy management. Section 6 highlights challenges and future trends. Finally, the conclusion, Section 7, summarizes the transformative role of cloud computing in IIoT.

Table 1. Summary of surveys and their descriptions.

Ref.	Description
[9]	Explores the integration of cloud computing and IoT with additive manufacturing, focusing on cloud platforms for resource sharing and scalable production in Industry 4.0. Discusses the challenges of combining legacy systems with state-of-the-art additive manufacturing technologies.
[10]	Surveys the application of blockchain technology in IIoT, emphasizing its role in enhancing security, traceability, and decentralization. Proposes blockchain for IIoT as a model for secure, interoperable, and autonomous industrial systems.
[11]	Provides a layered architecture for integrating blockchain into IIoT. Highlights blockchain's ability to secure devices, share data, and streamline industrial applications while addressing technical challenges like scalability and interoperability.
[12]	Examines how blockchain can address decentralization and trust issues in IoT/IIoT. Reviews smart contract applications in supply chains, energy, and healthcare and identifies future research directions.
[13]	Surveys IIoT from a control, networking, and computing perspective, detailing architectures for automation and interconnection in industrial systems. Focuses on edge computing and 5G integration for low-latency applications.
[14]	Discusses the role of edge computing in IIoT for reducing latency and optimizing bandwidth. Covers task scheduling, data analytics, and security challenges in deploying edge networks in industrial scenarios.
[15]	Analyzes the use of blockchain to address security and scalability challenges in IIoT. Focuses on decentralized architectures and applications like energy management, smart manufacturing, and healthcare data security.

**Figure 1.** An overview of the key discussed topics for applying cloud computing in IIoT.

2. Cloud Computing Architectures for IIoT

The foundation of effective IIoT deployments lies in selecting appropriate cloud computing architectures and service models. These architectures enable the processing, storage, and analysis of data generated by IIoT devices, ensuring that systems meet the demands of scalability, reliability, and efficiency. This section explores the key architectures—centralized and distributed—and the flexible service models that together form the backbone of IIoT systems.

2.1. Centralized Cloud Architectures for IIoT

Centralized cloud architectures play a pivotal role in IIoT environments by leveraging large-scale data centers for aggregating, storing, and processing vast amounts of data collected from distributed IIoT devices. These architectures provide a unified platform to enable global monitoring, predictive analytics, and optimization of industrial processes [16–18]. A key feature of centralized architectures is their scalability, which is achieved by dynamically allocating computational and storage resources to handle the ever-growing data streams generated by interconnected devices. If $D(t)$ represents the rate of data generation by N IIoT devices, it can be expressed as $D(t) = \sum_{i=1}^N d_i(t)$, where $d_i(t)$ is the data rate for device i . Centralized systems aggregate $D(t)$ in real time, facilitating the deployment of advanced analytics in industrial operations [19,20].

A central use of such architectures lies in predictive analytics. Let $\mathbf{X} \in \mathbb{R}^{m \times n}$ denote a dataset of m samples with n features stored in the cloud. Predictive models, such as regression or classification algorithms, operate on these data to minimize loss functions like the Mean Squared Error (MSE), defined as $\mathcal{L}(\mathbf{X}, \mathbf{y}) = \frac{1}{m} \sum_{i=1}^m (f(\mathbf{x}_i) - y_i)^2$, where $f: \mathbb{R}^n \rightarrow \mathbb{R}$ represents the predictive function, $\mathbf{x}_i$ denotes the feature vector i , and y_i is the target outcome of $\mathbf{x}_i$. Cloud infrastructure enables efficient computation of these loss functions by leveraging parallelized optimization techniques like gradient descent, thereby accelerating the training and deployment of ML models [21,22].

Another major advantage of centralized cloud architectures is their ability to implement digital twins, which are virtual models replicating physical industrial systems. These digital twins operate by continuously updating their state $\mathbf{S}(t)$ according to system dynamics, mathematically modeled as $\mathbf{S}(t+1) = \mathbf{F}(\mathbf{S}(t), \mathbf{u}(t))$, where $\mathbf{u}(t)$ denotes the control inputs, and $\mathbf{F}$ represents the dynamics of the physical system. By simulating various operational scenarios, digital twins provide real-time insights for predictive maintenance, fault detection, and optimization of manufacturing processes [23,24].

However, centralized architectures face significant challenges, particularly in latency-sensitive IIoT applications. The delay Δt incurred in such systems is composed of multiple factors, including data transmission time t_{transmit} , processing time t_{process} , and the time required to return a response to the source t_{response} . For industrial applications requiring ultra-low latency, such as robotic control or real-time quality inspection, this delay must not exceed a critical threshold τ . When $\Delta t > \tau$, centralized systems may fail to meet the stringent requirements of these applications [25,26].

Despite these limitations, centralized cloud architectures remain a cornerstone for applications such as predictive maintenance, where extensive historical data are analyzed to anticipate equipment failures, and quality assurance, where ML models detect defects in manufactured products with high precision. To address the challenges of latency and bandwidth constraints, hybrid approaches combining centralized cloud computing with edge computing are emerging as a promising solution. These hybrid systems strike a balance by performing local real-time processing at the edge while leveraging the cloud for global data aggregation and advanced analytics, ensuring that industries can meet both computational and responsiveness requirements effectively [27,28].

2.2. Distributed Cloud Architectures for IIoT

Distributed cloud architectures have emerged as a transformative approach to address the limitations of centralized cloud systems, particularly for latency-sensitive IIoT applications. By dispersing computational resources across multiple geographically distributed nodes, these architectures minimize latency, enhance fault tolerance, and provide localized data processing capabilities. The distributed nature of this approach allows for the pro-

cessing of IIoT data closer to the source, reducing the dependency on a single central data center and optimizing overall system efficiency [29–31].

In distributed architectures, device-generated data are processed at edge nodes or intermediate layers before being transmitted to a global data center for further analysis. If M distributed nodes are deployed, the data processed locally at each node j can be expressed as $P_j(t) = \alpha_j \cdot D(t)$, where $D(t)$ represents the total data rate by all devices in the system, and α_j represents the allocation fraction for node j such that $\sum_{j=1}^M \alpha_j = 1$. This allocation ensures that the workload is dynamically distributed across the network to prevent overload and optimize resource utilization [32–34].

A key advantage of distributed architectures is their ability to reduce latency. However, the extent of latency reduction depends on system design, as poorly optimized architectures may introduce delays from inter-node communication, synchronization, or resource contention. Achieving maximum benefits requires careful balancing of processing and communication overheads. The end-to-end delay Δt in such systems can be expressed as $\Delta t = t_{local} + t_{transmit} + t_{global}$, where t_{local} is the processing time at the local node, $t_{transmit}$ is the transmission time to a centralized cloud (if necessary), and t_{global} is the processing time at the global data center. By minimizing $t_{transmit}$ through local or regional data processing, distributed systems significantly reduce Δt , making them suitable for real-time applications such as industrial control, autonomous vehicles, and predictive maintenance [35,36].

Distributed cloud architectures also enhance fault tolerance. If a node fails, the system redistributes its workload among other nodes. Let $F_j(t)$ denote the failure state of node j , where $F_j(t) = 1$ if the node fails at time t , and $F_j(t) = 0$ otherwise. The redistributed workload for node k can then be expressed as $P'_k(t) = P_k(t) + \beta_j \cdot P_j(t)$, where β_j is the redistribution coefficient for the failed node j , ensuring that the system maintains operational continuity [37,38].

Applications of distributed cloud architectures include localized analytics for IIoT systems, such as real-time monitoring and anomaly detection. For instance, in a manufacturing setting, edge nodes process sensor data to identify anomalies in production lines, triggering immediate corrective actions. Moreover, distributed systems support hierarchical data aggregation, where local nodes perform preliminary analysis and transmit summarized results to global centers for comprehensive analytics, thereby reducing communication overheads [39,40].

Despite these advantages, distributed cloud architectures face challenges such as network synchronization, resource heterogeneity, and dynamic workload balancing. Addressing these issues requires sophisticated algorithms for resource management and adaptive scheduling. For example, distributed ML models, such as federated learning, optimize system-wide performance by training local models on edge nodes and aggregating their parameters centrally, reducing the need for raw data transmission while preserving data privacy [41–43].

2.3. Service Models in Cloud Computing for IIoT

Cloud computing service models—IaaS, PaaS, and SaaS—are foundational for the effective deployment of IIoT systems. These models provide varying levels of abstraction, enabling industries to tailor solutions to their specific operational needs while balancing customization, scalability, and efficiency [44,45].

IaaS offers fundamental computing resources such as virtual machines, storage, and networking infrastructure. For IIoT, this model provides industries with the flexibility to host and manage customized applications tailored to their operational needs. For instance, consider an IIoT system with N devices generating data at a rate of $d_i(t)$ and a collective rate of $D(t)$. An IaaS platform dynamically allocates computational resources

$R(t)$ proportional to $D(t)$, optimizing costs while ensuring high performance during peak workloads. This flexibility is particularly beneficial for processing-intensive tasks, such as real-time monitoring and predictive maintenance, where latency and performance are critical [46–48].

PaaS abstracts the underlying infrastructure, providing a development environment with pre-configured tools and frameworks. This model accelerates the creation of IIoT applications by enabling developers to focus solely on software development rather than hardware management. For example, PaaS supports the training of ML models for IIoT data analysis by providing libraries and computing resources to optimize a loss function $\mathcal{L}(\mathbf{X}, \mathbf{y})$ as defined in Section 2.1. This simplification of the development process allows industries to deploy applications for real-time analytics or anomaly detection in IIoT systems more efficiently [49–51].

SaaS provides ready-to-use cloud-hosted applications, eliminating the need for on-premises installations. Industries benefit from immediate access to solutions for asset monitoring, predictive maintenance, and process optimization. For instance, SaaS platforms provide tools for visualizing IIoT data in real time, such as dashboards displaying a set of metrics $\{m_k(t)\}_{k=1}^K$ that correspond to K monitored parameters. These dashboards enable operational decision-making by aggregating and analyzing IIoT data streams [52,53].

The versatility of these service models enables industries to align cloud computing adoption with their strategic goals. IaaS supports fully customizable environments, PaaS accelerates innovation by simplifying application development, and SaaS offers out-of-the-box solutions for immediate integration. By leveraging these models, industries can optimize operations, reduce costs, and enhance competitiveness in the evolving industrial landscape [54,55].

To provide further insights into the aforementioned concepts, Table 2 classifies key references that elaborate on the foundational principles, technological advancements, and practical implementations of these cloud computing models in IIoT environments. This classification serves as a guide for readers to explore specific aspects of cloud adoption in industrial settings, facilitating a deeper understanding of the trade-offs and challenges associated with different deployment strategies.

Table 2. Summary of works focusing on different architectural approaches, methodologies, concepts, and applications in IIoT.

References	Description	Key Concept	Applications	Challenges
[16–18]	Discuss large-scale data centers for aggregating, storing, and processing data collected from IIoT devices.	Scalability of centralized architectures	Global monitoring and predictive analytics	High latency and dependency on networks
[19,20]	Highlight real-time data aggregation for deploying advanced analytics on industrial operations.	Real-time aggregation	Predictive maintenance	Handling high data rates effectively
[21,22]	Emphasize the use of ML models stored in centralized clouds for reducing MSE during predictive analysis.	ML and loss minimization	Quality assurance in production	Computational load of ML optimization
[23,24]	Introduce digital twins that replicate physical systems using mathematical state transitions.	Digital twins for simulation	Fault detection, optimization	Complexity in maintaining real-time states
[25,26]	Examine latency issues in centralized systems for time-critical IIoT applications.	Latency and critical thresholds	Robotic control, quality inspection	Failing to meet ultra-low latency needs
[27,28]	Propose hybrid architectures combining centralized clouds with edge computing for better responsiveness.	Hybrid approaches	Real-time local processing with global analytics	Balancing computational distribution
[29–31]	Evaluate energy efficiency in centralized cloud architectures to optimize resource utilization in IIoT systems.	Energy efficiency	Cost reduction in operations	Energy overhead during peak loads
[32–34]	Discuss centralized management of IIoT networks for enhanced data security and privacy.	Centralized security management	Securing sensitive IIoT data	Risk of a single point of failure
[35,36]	Highlight data standardization issues in centralized architectures for cross-platform IIoT systems.	Data standardization	Integration of heterogeneous devices	Ensuring compatibility and interoperability

Table 2. Cont.

References	Description	Key Concept	Applications	Challenges
[37,38]	Describe challenges in bandwidth utilization when transmitting large-scale IIoT data to centralized systems.	Bandwidth optimization	Efficient data transfer	Bottlenecks during high traffic
[39,40]	Explore reliability enhancements in centralized architectures using redundant data centers.	Reliability improvements	Fail-safe mechanisms for critical systems	Costs of redundancy
[41–43]	Discuss cloud-based frameworks for predictive failure analysis and maintenance in industrial setups.	Predictive frameworks	Anomaly detection and fault prevention	High computation and storage requirements
[44,45]	Review scalability improvements in centralized systems for dynamic IIoT workloads.	Dynamic scalability	Handling spikes workload	Resource allocation challenges
[46–48]	Study centralized architectures' performance under real-time industrial scenarios.	Real-time performance	Automated industrial operations	Latency sensitivity
[49–51]	Propose methods to integrate centralized clouds with AI tools for advanced analytics in IIoT.	AI integration	Smart decision-making systems	Complexity in AI-cloud deployment
[52,53]	Describe compliance challenges in centralized systems for industrial standards and regulations.	Regulatory compliance	Meeting industrial standards	Ensuring adherence to evolving regulations
[54,55]	Highlight environmental impacts of centralized data centers and sustainability measures.	Sustainability measures	Eco-friendly IIoT operations	Balancing sustainability with performance

2.4. Comparative Analysis of Cloud Computing Architectures and Service Models for IIoT

Cloud computing architectures and service models play a crucial role in shaping the performance, scalability, and efficiency of IIoT deployments. Selecting the appropriate cloud architecture depends on several factors, including latency requirements, fault tolerance, and scalability. Centralized cloud architectures offer robust data aggregation and global resource management, making them suitable for applications that require large-scale data processing and long-term storage. However, they often introduce higher latency and dependency on network stability, which may not be ideal for real-time IIoT applications. In contrast, distributed cloud architectures enhance real-time processing by leveraging edge computing and localized data centers, reducing response times and improving system resilience. However, they introduce challenges in synchronization, workload balancing, and maintaining data consistency across multiple nodes. Hybrid approaches attempt to balance these trade-offs by combining the strengths of both models, enabling industries to optimize computational efficiency while minimizing latency concerns.

Similarly, the three primary cloud service models—Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS)—offer varying levels of flexibility and abstraction. IaaS provides direct control over computational resources, allowing industries to build customized IIoT solutions tailored to their specific needs. This model is beneficial for organizations that require specialized configurations but demands significant technical expertise for infrastructure management. PaaS abstracts infrastructure complexities, providing a development-friendly environment that simplifies the deployment of IIoT applications while offering pre-configured frameworks and tools. However, it may limit customization options. SaaS, in contrast, enables organizations to adopt cloud-based IIoT applications with minimal management overheads, offering ease of use and rapid deployment. While SaaS reduces infrastructure maintenance efforts, it may impose constraints on flexibility and data control, which could be a concern for industries requiring extensive customization.

Each of these models and architectures presents unique challenges. Centralized cloud solutions may face high data transmission costs and bottlenecks in latency-sensitive applications, while distributed models require sophisticated coordination to maintain consistency and interoperability. Similarly, while IaaS offers a high degree of customization, it requires significant expertise to manage, whereas PaaS and SaaS provide streamlined solutions with varying levels of control and adaptability. These trade-offs must be carefully

evaluated based on industry-specific requirements, operational constraints, and long-term scalability objectives.

All these aspects (definitions, latency characteristics, fault tolerance, scalability, resource allocation, challenges, and applications) are summarized in Table 3, providing a structured comparison to guide decision-makers in selecting the most suitable cloud computing model for IIoT deployments.

Table 3. Comparative summary of cloud computing architectures and service models for IIoT.

Aspect	Centralized	Distributed	Service Models
Definition	Aggregates all IIoT data in large, centralized data centers for global processing and storage.	Distributes computational and storage resources across geographically dispersed nodes to reduce latency and enhance fault tolerance.	Categorized into three models: IaaS (infrastructure provisioning), PaaS (development platforms), and SaaS (ready-to-use applications).
Latency	Higher latency due to the transmission of data to distant data centers.	Lower latency as processing occurs closer to the source of data generation.	Varies depending on the model. IaaS offers control at the infrastructure level, PaaS simplifies development, and SaaS provides instant solutions.
Fault Tolerance	Limited; failure of the central data center impacts the entire system.	High; localized nodes can redistribute workloads to maintain operational continuity in case of node failure.	IaaS and PaaS rely on redundancy at the infrastructure level; SaaS offers inherent fault tolerance managed by service providers.
Scalability	Supports global scalability but requires significant infrastructure investment for increasing capacity.	Scales efficiently through dynamic workload distribution across multiple nodes.	Highly scalable depending on the model. IaaS and PaaS enable on-demand resource scaling; SaaS scales seamlessly with additional users or devices.
Resource Allocation	Centralized allocation of resources for IIoT data processing and analytics.	Decentralized, with resources dynamically distributed to nodes based on local demand.	IaaS allows user-defined resource allocation; PaaS simplifies allocation for developers; SaaS abstracts resource management entirely.
Challenges	High latency, single point of failure, and dependency on robust network connections.	Complex synchronization, resource heterogeneity, and dynamic workload balancing.	Dependent on model-specific challenges: IaaS (complex management), PaaS (limited customization), and SaaS (less control over operations).
Application Example	Predictive maintenance using large-scale historical data for analysis and anomaly detection.	Real-time monitoring of manufacturing processes with edge-based processing for immediate responses.	IaaS supports customizable IIoT systems, PaaS accelerates ML development, and SaaS provides tools like dashboards for asset monitoring.

3. Communication and Data Management

Effective communication and data management are pivotal in enabling IIoT systems to operate seamlessly and achieve their objectives. Communication protocols ensure reliable and efficient data exchange between interconnected devices, while robust data storage and processing frameworks handle the massive and diverse data streams generated by these systems. Advanced integration of AI and ML into data workflows enhances these operations by enabling predictive analytics, anomaly detection, and autonomous decision-making, allowing for dynamic optimization of resources and improved system reliability. These components form the foundation for real-time analytics, decision-making, and system optimization in IIoT. This section explores the key aspects of communication protocols and data management, highlighting their significance in modern industrial environments.

3.1. Communication Protocols

Communication protocols are crucial in facilitating data exchange within IIoT systems, where diverse devices must interconnect and transmit information effectively. These protocols are designed to meet the unique requirements of IIoT, including low latency, energy efficiency, and secure transmission. Standard protocols such as MQTT (Message Queuing Telemetry Transport), CoAP (Constrained Application Protocol), and AMQP (Advanced Message Queuing Protocol) are widely adopted in IIoT environments due to their reliability and lightweight architecture, which are critical for resource-constrained devices [56–58].

MQTT, a lightweight publish/subscribe protocol, is particularly well suited for IIoT scenarios that involve frequent and asynchronous communication. Its low overhead enables efficient data transmission between devices with limited processing capabilities,

such as sensors and actuators in industrial plants. CoAP, on the other hand, is optimized for constrained devices and networks, making it an ideal choice for IIoT applications where bandwidth and power are limited. AMQP provides advanced messaging features, ensuring reliable delivery and enabling the integration of complex systems within IIoT environments [59–62].

OPC-UA (Open Platform Communications Unified Architecture) has emerged as a de facto standard for secure, platform-independent communication in IIoT. It supports hierarchical data structuring and seamless interoperability among diverse devices, addressing the need for robust industrial automation protocols [63,64].

The choice of communication protocol significantly impacts the overall performance and efficiency of IIoT systems. Protocols must strike a balance between speed, reliability, and resource consumption to ensure that devices can exchange data seamlessly without compromising system performance. As IIoT ecosystems grow more complex, the interoperability of these protocols becomes critical to ensure smooth communication between heterogeneous devices and platforms. This interoperability fosters the integration of legacy systems with modern IIoT frameworks, ensuring that industrial operations remain adaptable and future-proof [65–67].

Emerging technologies such as 5G and software-defined networking (SDN) are transforming the IIoT communication landscape. The 5G network provides ultra-low latency and high bandwidth, while SDN enables programmable networks to adapt dynamically to changing demands [68,69]. Network Function Virtualization (NFV) acts as a key enabler, deploying virtualized network functions (VNF) to meet Quality of Service (QoS) and low-latency requirements. Ensuring latency lower than a threshold τ (for satisfying the required QoS demands) is critical for real-time applications like industrial control systems [70–72].

3.2. Data Storage and Processing

Data storage and processing are at the heart of IIoT systems, providing the infrastructure needed to handle the vast and diverse data generated by connected devices. These systems produce data in various forms, such as sensor readings, machine logs, and real-time analytics streams, requiring robust solutions to store and process information efficiently. Cloud-based storage frameworks have become the cornerstone of IIoT data management, offering scalability, flexibility, and cost efficiency [73–75].

Cloud storage enables IIoT systems to centralize data from geographically dispersed devices, providing a unified platform for analysis and decision-making. By leveraging distributed cloud architectures, industries can process data locally at edge nodes or transmit them to centralized servers for comprehensive analytics. This hybrid approach ensures that latency-sensitive applications benefit from real-time processing while historical data are stored for long-term analysis and ML model training [76–78].

Kubernetes, the open-source de facto standard for managing containerized workloads and services, has revolutionized data management in IIoT [79]. **Kubernetes automates scaling, load balancing, and resource allocation**, ensuring high availability and fault tolerance. These features are particularly beneficial in dynamic IIoT environments, where workloads and resource demands can fluctuate significantly. Kubernetes' ability to distribute workloads intelligently across nodes enhances system performance and reliability, making it a cornerstone technology for modern IIoT applications [80–82].

Advanced data processing techniques, such as stream processing and batch analytics, are integral to extracting actionable insights from IIoT data. Stream processing frameworks, such as Apache Kafka and Apache Flink, allow real-time analysis of data as they are generated, enabling immediate responses to events such as equipment failures or environmental

changes. Batch processing systems, such as Hadoop and Spark, are used for analyzing large datasets over longer periods, identifying trends, and optimizing industrial operations [83–87].

Blockchain technologies are also gaining traction in IIoT for ensuring data integrity and tamper-proof storage. By maintaining an immutable ledger, blockchain enhances trust in data sharing, particularly for compliance with regulatory standards [88].

3.3. Integration of AI and ML into Data Workflows

As IIoT systems continue to evolve, the demand for scalable, secure, and intelligent data storage and processing solutions will only grow. The integration of AI and ML into data management workflows is driving the development of predictive analytics, anomaly detection, and autonomous decision-making capabilities. These advancements are empowering industries to harness the full potential of their data, transforming raw information into actionable intelligence and enabling a new era of industrial innovation [89–92].

The integration of AI and ML into IIoT data workflows has revolutionized predictive and preventive analytics by leveraging real-time data streams and historical datasets [93]. By employing advanced ML algorithms such as random forests, support vector machines, or deep learning (DL) architectures like convolutional neural networks (CNNs) and long short-term memory (LSTM) networks, industries can detect complex patterns in data that human operators might overlook [94,95]. For instance, in manufacturing, vibration and acoustic sensor data collected from machinery can be analyzed to identify wear patterns or predict failures before they occur. Mathematically, anomaly detection often involves comparing real-time data $x_i(t)$ to an expected distribution $P(X)$, with deviations quantified through statistical measures like Mahalanobis distance or loss functions minimized during model training [96,97].

Anomaly detection in IIoT systems benefits immensely from AI's capability to classify and prioritize alerts with contextual relevance. Autoencoders and generative adversarial networks (GANs) are increasingly deployed to reconstruct normal operational states and flag deviations as anomalies. This is particularly valuable in critical systems such as chemical plants or power grids, where anomalies can signal catastrophic failures [98–100]. Techniques such as Bayesian optimization or reinforcement learning (RL) further enhance decision-making by dynamically adapting the anomaly thresholds θ based on evolving environmental conditions. These frameworks not only improve detection accuracy but also reduce the likelihood of false positives, ensuring operational reliability [101,102].

AI-driven autonomous decision-making in IIoT employs RL models like Deep Q-Networks (DQN) and actor-critic algorithms, which learn optimal actions through continuous interaction with the environment. For example, an industrial robotic arm integrated with IIoT sensors can adjust its operational parameters based on feedback loops, maximizing throughput while minimizing energy consumption. RL models learn decision policies $\pi(a|s)$, which guide the system's choice of actions a based on the current state s by maximizing cumulative rewards of the form $R_t = \sum_{i=0}^{\infty} \gamma^i r_{t+i}$, where r_t represents the immediate benefit of action a at time t and γ is the discount factor prioritizing short-term or long-term rewards. In IIoT, this framework supports autonomous decision-making for tasks like optimizing resource allocation, scheduling predictive maintenance, and minimizing energy usage. These methods ensure systems can autonomously adapt to changing conditions, enhancing reliability and efficiency and supporting complex industrial operations without requiring constant human oversight [103–105].

In summary, integrating AI and ML into IIoT workflows has enabled smarter predictive analytics, precise anomaly detection, and autonomous decision-making, transforming industrial operations. Advanced algorithms and real-time data processing ensure these systems adapt dynamically, enhancing efficiency and reliability. As these technologies

evolve, they promise even greater autonomy and operational excellence across diverse industrial applications. Table 4 details key topics such as communication protocols, data storage, AI integration, and advanced technologies. It emphasizes their roles in enhancing IIoT systems through scalability, security, and real-time data processing. It also highlights core concepts and their applications in industrial environments.

Table 4. Communication protocols, data storage, AI integration, and advanced technologies in IIoT.

References	Description	Key Concept	Application/Insight
[56–58]	Explore MQTT, CoAP, and AMQP protocols emphasizing lightweight and efficient data exchange for IIoT.	Lightweight communication protocols for IIoT.	Enables efficient and low-power communication in constrained IIoT devices.
[59–62]	Detailed analyses of messaging protocols ensuring reliable delivery and handling complex IIoT messaging scenarios.	Reliable and advanced messaging protocols.	Supports reliable data exchange in complex industrial environments.
[63,64]	Focus on OPC-UA as a secure, platform-independent protocol facilitating robust IIoT communication.	Secure and standardized communication architecture.	Facilitates secure and seamless communication among diverse industrial devices.
[65–67]	Examine interoperability challenges in IIoT communication systems for integrating legacy and modern platforms.	Interoperability across heterogeneous systems.	Addresses integration of diverse devices and legacy systems in IIoT frameworks.
[68,69]	Highlight the potential of 5G and SDN technologies for improving IIoT communication efficiency.	Technologies enhancing IIoT communication efficiency.	Drives ultra-low latency and high-bandwidth solutions for industrial IoT applications.
[70–72]	Analyze NFV's role in enabling QoS and reducing communication latency in IIoT applications.	NFV enables low latency and high QoS in IIoT.	Optimizes network resource allocation to meet stringent IIoT requirements.
[73–75]	Cloud-based storage frameworks for centralized data management in IIoT systems.	Centralized data storage for IIoT.	Enhances unified data management and scalability.
[76–78]	Hybrid cloud and edge computing for balancing real-time processing and historical data analysis.	Hybrid architectures combining cloud and edge processing.	Optimizes latency-sensitive and historical data operations.
[79–82]	Kubernetes as a tool for automating resource allocation and management in IIoT environments.	Resource automation and management via Kubernetes.	Improves fault tolerance and resource utilization in IIoT systems.
[83–87]	Real-time data analytics using stream and batch processing frameworks like Kafka and Hadoop.	Real-time and batch data analytics in IIoT.	Enables timely and actionable insights for industrial applications.
[88]	Blockchain for ensuring data integrity and tamper-proof storage in IIoT systems.	Blockchain for secure and immutable data handling.	Ensures compliance and trust in data sharing across platforms.
[89–93]	AI and ML integration for enhancing predictive and anomaly detection capabilities in IIoT workflows.	AI/ML for predictive analytics and anomaly detection.	Improves prediction accuracy and operational efficiency.
[94,95]	Application of DL architectures like CNNs and LSTMs for complex pattern recognition.	DL for advanced IIoT applications.	Identifies patterns in IIoT data that are challenging for traditional methods.

Table 4. Cont.

References	Description	Key Concept		Application/Insight
[96,97]	Use of statistical measures for anomaly detection in IIoT systems.	Statistical measures for detecting anomalies.		Quantifies and identifies deviations from expected IIoT behavior.
[98–100]	Autoencoders and GANs for anomaly detection in critical industrial systems.	Advanced detection	anomaly using generative models.	Detects critical system failures with high precision.
[101,102]	Dynamic threshold adaptation using Bayesian optimization and RL for improved reliability.	Adaptive thresholds for operational reliability.	anomaly for operational reliability.	Enhances accuracy and reduces false positives in anomaly detection.
[103–105]	RL models like DQN for autonomous decision-making in industrial systems.	Autonomous industrial decision-making using RL models.		Optimizes industrial processes and reduces human intervention.

4. Security and Privacy in IIoT Cloud Integration

The integration of cloud computing with IIoT has unlocked transformative capabilities in industrial operations, but it has also introduced critical challenges in security and privacy. IIoT systems often involve sensitive data, such as operational metrics, customer information, and proprietary industrial processes, making them attractive targets for cyber-attacks. The dynamic nature of IIoT, coupled with the need for seamless connectivity between devices and cloud platforms, adds further complexity to ensuring secure and private operations. This section explores the key challenges and innovative solutions for safeguarding IIoT cloud ecosystems.

4.1. Security and Privacy Challenges

Security and privacy in IIoT–cloud integration face numerous challenges, primarily due to these systems’ distributed and heterogeneous nature. One of the foremost concerns is the vulnerability of IIoT devices, which are often resource-constrained and lack robust security mechanisms. Many devices operate with limited computational and power capabilities, making it challenging to implement traditional security protocols without affecting performance. As a result, these devices become potential entry points for attackers aiming to compromise the broader network [106–108].

IoT device tampering poses significant risks to the integrity of IIoT systems. Attackers can manipulate device firmware or software to introduce malicious code, leading to network poisoning or unauthorized access to critical systems. Such tampering not only disrupts device functionality, but also compromises the security of interconnected networks, enabling attackers to propagate malware or exfiltrate sensitive data. Addressing device tampering requires secure boot mechanisms and hardware trust anchors to ensure that devices operate with verified and untampered firmware [109–111].

Data in transit and at rest also present significant security challenges. IIoT systems rely on continuous data transmission between devices and cloud platforms, often over public or semi-private networks. This opens the door to risks such as man-in-the-middle attacks, data interception, and unauthorized access. Ensuring secure communication requires robust encryption and authentication protocols, but implementing these consistently across diverse devices and platforms can be challenging. Furthermore, storing large volumes of industrial data in cloud environments raises concerns about data breaches and unauthorized access to sensitive information [112–115].

Data sovereignty is another critical concern in IIoT systems, particularly in industries operating across multiple jurisdictions. Data sovereignty refers to the principle that data are subject to the laws and governance structures of the country where they are collected

or stored. Non-compliance with sovereignty requirements can result in legal penalties and compromise operational integrity. Ensuring correct management and control of data ownership is essential for maintaining compliance with regulations and ensuring the trust of industrial partners [116–118].

Another critical challenge lies in maintaining privacy and compliance in IIoT systems. Industries must adhere to regulations such as GDPR (General Data Protection Regulation), CCPA (the California Consumer Privacy Act), and other data protection laws, which impose strict requirements on how data are collected, stored, and processed. Ensuring compliance becomes particularly complex when IIoT deployments span multiple jurisdictions with varying regulatory frameworks. Additionally, balancing the need for data sharing in collaborative industrial ecosystems with privacy protection adds another layer of complexity [119–121].

The dynamic and interconnected nature of IIoT systems further exacerbates security challenges. With thousands of devices operating in real time, maintaining system integrity and preventing unauthorized access becomes a monumental task. Attackers can exploit vulnerabilities in one device to gain access to the entire network, resulting in cascading effects that disrupt industrial operations. These challenges highlight the pressing need for innovative and scalable security solutions tailored to the unique requirements of IIoT–cloud integration [122–125].

4.2. Solutions

Industries are adopting innovative and multi-layered solutions to address the multifaceted security and privacy challenges of IIoT–cloud integration. One foundational approach involves strengthening device security by incorporating secure boot mechanisms, hardware-based encryption, and lightweight cryptographic protocols. These measures ensure that IIoT devices can defend against common threats while maintaining optimal performance. The use of trusted hardware modules and secure firmware updates further enhances the resilience of connected devices [126–129].

Data protection is another critical area where advanced solutions are making a significant impact. End-to-end encryption ensures that data remain secure during transmission between devices and cloud platforms, minimizing the risk of interception. Cloud providers are increasingly adopting zero-trust architectures, where every access request is authenticated and authorized, regardless of its origin. This model significantly reduces the likelihood of unauthorized access to sensitive data stored in cloud environments [130–134].

Blockchain technology is emerging as a game-changer in securing IIoT–cloud systems. By creating immutable and transparent records of data exchanges, blockchain enhances trust and data integrity within IIoT networks. For example, blockchain-based smart contracts can automate access control policies, ensuring that only authorized entities can interact with specific devices or datasets. Additionally, the decentralized nature of blockchain eliminates single points of failure, enhancing the overall robustness of IIoT ecosystems [135–138].

Privacy-preserving techniques such as differential privacy and federated learning are addressing the dual challenge of data sharing and privacy protection. Differential privacy adds noise to datasets, ensuring that individual data points cannot be identified, even in aggregate analyses. Federated learning enables devices to collaboratively train ML models without sharing raw data, preserving privacy while extracting value from distributed data sources. These methods are particularly effective in industries where data sovereignty and compliance are critical considerations [139–142].

Finally, real-time threat detection and response mechanisms are becoming integral to IIoT security strategies. AI-driven intrusion detection systems can analyze network traffic patterns, identify anomalies, and respond to potential threats in real time. These systems

adapt and evolve with the threat landscape, providing proactive defense against emerging vulnerabilities. By combining advanced security technologies with rigorous compliance practices, industries can create resilient and secure IIoT-cloud environments that support innovation while safeguarding critical assets [143–145].

Addressing security and privacy in IIoT-cloud integration involves tackling challenges such as device tampering, secure communication, data sovereignty, and regulatory compliance. Proposed solutions include hardware-based security, end-to-end encryption, zero-trust architectures, and blockchain for data integrity and access control. Privacy-preserving techniques like federated learning and differential privacy help balance data sharing with confidentiality, while AI-driven intrusion detection systems enhance real-time threat detection and responses. Table 5 summarizes these challenges and solutions, categorizing key studies based on focus areas such as device-level security, secure communication, system integrity, compliance, and AI-driven threat detection. This classification provides a structured reference for understanding existing security strategies and identifying future research directions.

Table 5. Summary of works addressing challenges and solutions in IIoT security.

References	Description	Key Concept
[106–108]	Address the challenges of IIoT device tampering and the importance of secure boot mechanisms and hardware trust.	IIoT device security and tampering prevention
[109–111]	Focus on ensuring secure communication with encryption and authentication protocols in IIoT systems.	Secure communication in IIoT systems
[112–115]	Highlight the significance of data sovereignty in IIoT systems, emphasizing compliance with local regulations.	Data sovereignty and compliance in IIoT
[116–118]	Discuss maintaining privacy and compliance in IIoT systems across multiple jurisdictions.	Privacy challenges in global IIoT deployments
[119–121]	Explore the dynamic and interconnected nature of IIoT and its impact on security and system integrity.	System integrity in dynamic IIoT environments
[122–125]	Investigate scalable and innovative solutions for the security challenges in IIoT-cloud integration.	Scalable security solutions for IIoT-cloud integration
[126–129]	Examine secure boot mechanisms and hardware encryption to enhance IIoT device security.	Hardware-based security for IIoT devices
[130–134]	Discuss the adoption of zero-trust architectures for improving IIoT data security and privacy.	Zero-trust security models in IIoT
[135–138]	Explore blockchain-based solutions for ensuring data integrity and automating access control in IIoT.	Blockchain for data integrity and access control
[139–142]	Highlight privacy-preserving techniques like differential privacy and federated learning in IIoT.	Privacy-preserving techniques in IIoT
[143–145]	Focus on AI-driven intrusion detection systems for real-time threat detection and responses in IIoT.	AI-driven intrusion detection in IIoT

5. Applications of Cloud Computing in IIoT

The integration of cloud computing into IIoT ecosystems has transformed how industries operate, unlocking opportunities to optimize processes, reduce costs, and enhance decision-making. Cloud platforms enable the aggregation, analysis, and sharing of data from diverse sources, providing a foundation for advanced applications across multiple

domains. This section explores key applications of cloud computing in IIoT, highlighting its role in predictive maintenance, industrial automation, supply chain optimization, and energy management.

5.1. Predictive Maintenance

Predictive maintenance has become one of the most impactful applications of cloud computing in IIoT, allowing industries to shift from reactive to proactive maintenance strategies. By leveraging cloud-based analytics and ML models, IIoT systems can monitor the health of equipment in real time and predict potential failures before they occur. Data from sensors embedded in machines, such as temperature, vibration, and pressure readings, are continuously transmitted to the cloud, where they are analyzed to detect anomalies and trends indicative of wear or malfunction [146–149].

The scalability of cloud computing enables predictive maintenance systems to process vast amounts of sensor data from multiple assets simultaneously, regardless of their geographic location. Advanced algorithms trained on historical and real-time data can provide accurate predictions of failure modes and suggest optimal maintenance schedules. This minimizes unplanned downtime, extends the lifespan of equipment, and reduces maintenance costs by avoiding unnecessary repairs [150–152].

Industries such as manufacturing, transportation, and energy have widely adopted predictive maintenance to enhance operational reliability. For example, in the manufacturing sector, cloud-integrated IIoT systems can monitor the performance of critical machinery, ensuring production lines operate without interruptions. Similarly, energy providers can predict the maintenance needs of wind turbines or power transformers, optimizing resource allocation and reducing service disruptions [153–155].

By integrating cloud computing with IIoT for predictive maintenance, industries gain a powerful tool for enhancing operational efficiency and decision-making. The ability to act on predictive insights ensures that resources are used effectively and systems are maintained in peak condition, driving significant improvements in productivity and cost savings [156,157].

5.2. Industrial Automation

Industrial automation represents a cornerstone of modern IIoT applications, where cloud computing plays a pivotal role in enabling autonomous and intelligent operations. Industries can streamline workflows, enhance precision, and achieve real-time responsiveness by connecting machines, robots, and control systems to the cloud. Cloud platforms facilitate centralized monitoring and control of automated processes, allowing industries to optimize production lines, improve quality assurance, and reduce human intervention [158–161].

One key advantage of cloud-integrated industrial automation is the ability to process and analyze large volumes of data in real-time. For instance, sensor data from automated machines can be transmitted to the cloud, where advanced analytics identify inefficiencies or deviations from standard operating conditions. These insights enable rapid adjustments, ensuring that processes remain efficient and meet desired quality standards [162–164].

Cloud-based automation also fosters collaboration between geographically dispersed facilities by providing a unified platform for managing operations. Industries can deploy and update automation protocols remotely, ensuring consistency across locations. Furthermore, cloud computing supports digital twins, virtual replicas of physical systems that allow industries to simulate and optimize automated workflows before implementation. This reduces the risk of errors and minimizes downtime during system changes [165–168].

Cloud-enabled automation has revolutionized operations in sectors such as automotive manufacturing, pharmaceuticals, and food processing. For example, robotic arms

on assembly lines can leverage cloud-based intelligence to adapt to changing production requirements, improving flexibility and productivity. By integrating cloud computing into industrial automation, industries can achieve higher levels of efficiency, accuracy, and scalability, driving competitiveness in an increasingly digitalized market [169–171].

5.3. Supply Chain Optimization

Supply chain optimization is a critical area where cloud computing and IIoT converge to enhance transparency, efficiency, and responsiveness. Cloud platforms enable end-to-end visibility across supply chain networks, allowing industries to track inventory, monitor logistics, and respond to disruptions in real-time. By integrating data from IIoT devices, such as Global Positioning System (GPS) trackers and Radio-Frequency Identification (RFID) tags, cloud-based systems provide actionable insights that drive smarter decision-making [172–175].

Cloud computing enhances supply chain management by enabling predictive analytics and demand forecasting. Historical sales data, weather patterns, and market trends can be analyzed to predict demand fluctuations, ensuring optimal inventory levels and reducing stockouts or overstock situations. Additionally, cloud platforms support route optimization for logistics, reducing transportation costs and improving delivery times by analyzing traffic, weather, and fleet performance data [176–178].

In manufacturing and retail, cloud-enabled IIoT systems streamline inventory management by providing real-time updates on stock levels, locations, and movement. For instance, warehouses equipped with IIoT sensors can automatically reorder supplies based on pre-defined thresholds, reducing manual intervention and errors. Similarly, cloud-based analytics can identify bottlenecks in supply chain workflows, enabling industries to implement targeted improvements [179–181].

The ability to integrate multiple data sources and provide a unified view of supply chain operations makes cloud computing an invaluable asset for industries seeking to remain competitive in a fast-paced global market. By leveraging IIoT and cloud technologies, supply chains become more agile, resilient, and cost-effective, ensuring that industries can adapt to changing demands and maintain customer satisfaction [182–185].

5.4. Energy Management

Energy management is another domain where cloud computing has made significant contributions to IIoT applications, enabling industries to optimize energy consumption, reduce costs, and meet sustainability goals. IIoT sensors embedded in industrial equipment and facilities continuously collect data on energy usage, which are transmitted to the cloud for analysis. Cloud-based systems process these data to identify inefficiencies, track trends, and recommend energy-saving measures [186–188].

At the sensor level, energy management is critical for ensuring the longevity and reliability of IIoT deployments. Many sensors operate on limited battery autonomy, making it essential to optimize their energy consumption. Cloud platforms can assist by scheduling sensor tasks on optimal nodes, balancing system constraints while preserving battery life. This approach enhances the operational efficiency of IIoT networks while minimizing downtime due to sensor failure [189–191].

One of the primary benefits of cloud-enabled energy management is the ability to implement real-time monitoring and control of energy consumption. For example, cloud platforms can integrate with building management systems to optimize heating, ventilation, and air conditioning (HVAC) operations based on occupancy and weather conditions. Similarly, manufacturing facilities can monitor the energy usage of individual machines,

ensuring that resources are allocated efficiently and unnecessary consumption is minimized [192–195].

Cloud computing also facilitates the integration of renewable energy sources into industrial operations. By leveraging IIoT and cloud platforms, industries can monitor the performance of solar panels, wind turbines, and other renewable assets, ensuring they operate at peak efficiency. Furthermore, cloud-based analytics can predict energy generation patterns and align them with consumption needs, reducing reliance on non-renewable sources and supporting sustainability initiatives [196–198].

The role of cloud computing in energy management extends to demand response programs, where industries adjust their energy usage in response to grid demands. Cloud platforms enable real-time communication with utilities, allowing industries to participate in programs that reduce peak load stress and earn financial incentives. This benefits the grid and aligns with industries' goals of reducing operational costs and environmental impact [199–201].

By integrating cloud computing with IIoT for energy management, industries gain a powerful tool for achieving energy efficiency and sustainability. The insights derived from cloud-based analytics enable smarter energy usage, supporting both economic and environmental objectives and driving progress toward a more sustainable industrial landscape [202,203].

To further illustrate these applications, Table 6 provides a structured summary of them, categorizing key domains, associated cloud-based technologies, and their benefits. This approach highlights the diverse ways in which cloud computing enhances IIoT systems, offering insights into its role in improving operational efficiency, cost reduction, and decision-making capabilities.

Table 6. Applications of cloud computing in IIoT.

References	Description	Key Application	Technology/Framework		Benefits
[146–149]	Explore predictive maintenance using real-time sensor data and cloud-based analytics.	Predictive Maintenance	Cloud Analytics		Reduces downtime and maintenance costs.
[150–152]	Highlight cloud-enabled fault detection systems in manufacturing equipment.	Fault Detection	Cloud-based Fault Detection		Enhances reliability and fault prediction accuracy.
[153–155]	Focus on the integration of cloud computing for remote diagnostics and maintenance.	Remote Diagnostics	Remote Frameworks	Monitoring	Improves diagnostic efficiency and accessibility.
[156,157]	Examine cost reduction strategies through proactive maintenance scheduling.	Cost Optimization	Proactive Algorithms	Scheduling	Minimizes unexpected breakdown costs.
[158–161]	Analyze cloud-based industrial automation for enhanced efficiency and precision.	Industrial Automation	Cloud-integrated Control Systems		Increases precision and workflow efficiency.
[162–164]	Explore the role of cloud platforms in centralized control for automation systems.	Centralized Automation	Cloud-based Tools	Automation	Improves consistency in automation protocols.
[165–168]	Highlight the use of digital twins for simulation and optimization in cloud-integrated IIoT.	Digital Twin Simulation	Digital Twin Platforms		Optimizes processes and reduces errors.
[169–171]	Discuss cloud-enabled adaptive automation protocols for industrial processes.	Adaptive Automation	Adaptive Cloud Protocols		Increases flexibility and adaptability.
[172–175]	Focus on supply chain optimization using real-time cloud-based data integration.	Supply Chain Optimization	Cloud-based Integration	Data	Improves supply chain transparency and decision-making.
[176–178]	Highlight route optimization and inventory tracking in cloud-based supply chains.	Route Optimization	Logistics Platforms		Reduces costs and improves logistics delivery efficiency.
[179–181]	Discuss predictive demand forecasting and logistics optimization with IIoT clouds.	Demand Forecasting	Predictive Analytics Frameworks		Enhances forecasting accuracy and planning.
[182–185]	Examine operational transparency and resilience enabled by cloud-based supply chains.	Operational Resilience	Cloud-based Transparency Solutions		Increases resilience to disruptions.
[186–188]	Explore energy management with IIoT cloud integration for monitoring and optimization.	Energy Monitoring	Energy Monitoring Tools		Optimizes energy use and reduces waste.

Table 6. Cont.

References	Description	Key Application	Technology/Framework	Benefits
[189–191]	Discuss battery optimization for IIoT sensors through cloud task scheduling.	Sensor Battery Optimization	Cloud Task Scheduling	Extends sensor lifespan and reduces operational costs.
[192–195]	Focus on HVAC optimization and energy consumption reduction using IIoT clouds.	Energy Consumption Reduction	HVAC Optimization Tools	Minimizes energy consumption in industrial facilities.
[196–198]	Examine renewable energy integration and management in cloud-enabled IIoT systems.	Renewable Energy Management	Renewable Energy Management Systems	Supports sustainable energy use and reduces dependency on non-renewable sources.
[199–201]	Discuss demand response programs for energy optimization in IIoT environments.	Demand Response Programs	Demand Response Platforms	Balances grid demands and reduces peak loads.
[202,203]	Highlight sustainability initiatives driven by cloud-enabled industrial energy management.	Sustainability in IIoT	Sustainability Tools	Aligns industrial operations with environmental goals.

6. Challenges and Future Trends

As cloud computing and IIoT continue to converge, the potential for industrial transformation is enormous. However, realizing this potential is fraught with challenges, ranging from technical and operational hurdles to concerns around security and sustainability. Addressing these challenges is essential to fully harnessing these technologies' capabilities. Concurrently, emerging trends in technology and industrial innovation offer promising avenues for overcoming limitations and expanding the scope of IIoT applications. This section examines both the challenges and the future trends shaping the trajectory of cloud computing in IIoT.

6.1. Challenges

The integration of cloud computing with IIoT faces numerous challenges that stem from the complexity and scale of these interconnected systems. A primary concern is the latency inherent in centralized cloud architectures, particularly for time-critical IIoT applications such as industrial automation and robotics. While cloud platforms excel in computational power and data storage, the reliance on remote data centers can lead to delays in processing and decision-making, potentially disrupting real-time operations. Addressing this issue requires the adoption of hybrid architectures that combine cloud computing with edge and fog computing to ensure local processing capabilities [204–208].

Another significant challenge is the sheer volume and diversity of data generated by IIoT devices. Managing and analyzing these data in real time requires highly scalable and efficient systems, which can strain existing cloud infrastructures. Moreover, the heterogeneity of IIoT devices and protocols adds complexity to data integration, interoperability, and standardization. Without unified standards, industries face difficulties in ensuring seamless communication and collaboration between devices and platforms, which can limit the efficiency and scalability of IIoT systems [209–212].

Containerization and microservices are emerging as critical solutions for managing the complexity and scalability of IIoT systems. By deploying lightweight software components, containerized applications can be efficiently orchestrated using Kubernetes, enabling dynamic scaling and optimized resource usage. These containers can also run on edge devices, enhancing local processing capabilities while maintaining seamless integration with cloud systems. This approach minimizes latency and improves overall system responsiveness, particularly for real-time industrial applications [213–216].

Serverless computing at the service level offers another innovative solution for optimizing cloud resource usage while meeting system requirements. Serverless architectures allow industries to execute specific tasks or services on-demand without managing the underlying infrastructure. This approach reduces operational costs and improves resource

efficiency, ensuring that computational resources are allocated only when needed, thereby aligning with the dynamic demands of IIoT applications [217–220].

Security and privacy concerns are critical challenges in IIoT–cloud integration. Industrial systems often involve sensitive operational data and intellectual property, making them prime targets for cyber-attacks. The distributed nature of IIoT further increases the attack surface, with vulnerabilities in individual devices potentially compromising entire networks. Ensuring robust security while maintaining performance and scalability is a complex balancing act, requiring advanced encryption, authentication, and monitoring solutions. Compliance with data protection regulations across jurisdictions adds another layer of complexity, as industries must navigate varying legal requirements while maintaining global operations [221–223].

Additionally, energy consumption and sustainability pose challenges for large-scale cloud-integrated IIoT systems. The computational and storage demands of IIoT result in significant energy usage, contributing to operational costs and environmental impact. Developing energy-efficient algorithms, hardware, and cloud architectures is crucial to addressing these concerns and ensuring the long-term viability of cloud-enabled IIoT systems [224–226].

These challenges underscore the complexity of integrating cloud computing with IIoT, particularly in environments requiring real-time analytics and high scalability. Table 7 provides a taxonomy of references discussed in this subsection, offering a detailed insight into the multifaceted challenges associated with real-time analytics in IIoT systems.

Table 7. Challenges in cloud computing for IIoT.

References	Description	Key Challenge	Impact on IIoT
[204–208]	Explore the challenge of ensuring data security and privacy in cloud-based IIoT systems.	Data Security and Privacy	Limits trust and adoption of cloud solutions in sensitive industrial applications.
[209–212]	Highlight latency issues in cloud-integrated IIoT environments due to centralized processing.	Latency in Cloud-IIoT	Reduces the performance of time-sensitive IIoT applications.
[213–216]	Discuss interoperability concerns between diverse IIoT devices and cloud platforms.	Interoperability	Hinders seamless integration and communication across devices and platforms.
[217–220]	Examine scalability limitations when managing large-scale IIoT deployments with cloud resources.	Scalability	Constrains the expansion and efficiency of IIoT systems.
[221–223]	Analyze the high energy consumption of cloud-based IIoT operations.	Energy Consumption	Increases operational costs and reduces sustainability.
[224–226]	Focus on the challenge of maintaining real-time service continuity during network disruptions.	Service Continuity	Disrupts real-time data processing and industrial operations.

6.2. Future Trends

Despite the challenges, future trends in technology and industrial practices promise to enhance the integration of cloud computing and IIoT, unlocking new possibilities for innovation and efficiency. One of the most significant trends is the increasing adoption of hybrid and multi-cloud architectures. By combining the strengths of centralized clouds with edge and fog computing, these architectures address latency and scalability issues while ensuring that data are processed and stored optimally. Industries are also exploring cloud-native solutions that leverage containerization and microservices to build flexible and resilient IIoT systems [227–230].

The proliferation of 5G and beyond will revolutionize IIoT communication, providing ultra-low latency, high bandwidth, and reliable connectivity. These advancements will enable seamless integration between IIoT devices and cloud platforms, facilitating real-time data exchange and decision-making. Additionally, 5G networks will support the

massive device density required for IIoT applications, further enhancing scalability and efficiency. Coupled with SDN and network slicing, these technologies will allow industries to customize network configurations to meet specific application requirements [231–234].

AI and ML are poised to play a transformative role in the future of cloud-enabled IIoT. AI-driven analytics will enhance predictive maintenance, optimize supply chains, and enable autonomous decision-making across industrial processes. Federated learning, in particular, offers a promising approach to AI in IIoT by allowing devices to collaboratively train models without sharing raw data, thus preserving privacy while improving accuracy. The integration of AI at the edge will further reduce latency and enable real-time insights, driving significant improvements in operational efficiency [235–237].

Sustainability will be a key focus in the future of cloud-enabled IIoT. Industries are increasingly prioritizing energy efficiency and environmental responsibility, driving the development of green computing technologies. Renewable-energy-powered data centers, energy-efficient hardware, and intelligent resource allocation algorithms are among the innovations shaping sustainable IIoT ecosystems. Additionally, circular economy principles are gaining traction, encouraging industries to adopt practices that minimize waste and maximize resource reuse [238–240].

In conclusion, while challenges such as latency, scalability, security, and sustainability remain significant, emerging trends in hybrid architectures, advanced communication technologies, AI-driven solutions, and sustainability initiatives offer a pathway to overcoming these barriers. The continued evolution of cloud computing and IIoT will empower industries to unlock new levels of innovation, efficiency, and resilience, shaping the future of industrial operations in the digital era [241,242]. Table 8 provides a comprehensive classification of references that highlight emerging trends in cloud-enabled IIoT systems. These trends underscore the importance of innovations in hybrid architectures, AI-driven analytics, and sustainable practices in shaping the future of industrial operations.

Table 8. Future Trends in IIoT.

References	Description	Emerging Trend	Technological Focus	Expected Impact
[227–230]	Explore the integration of edge computing with cloud systems to reduce latency and enhance processing efficiency.	Edge–Cloud Integration	Edge Computing, Cloud Systems	Improves real-time processing and reduces latency for IIoT applications.
[231–234]	Discuss the adoption of 5G technologies to improve network performance in IIoT environments.	5G Network Deployment	5G Infrastructure	Enhances connectivity, bandwidth, and reliability for industrial applications.
[235–237]	Analyze the role of federated learning for privacy-preserving AI in distributed IIoT systems.	Privacy-Preserving AI	Federated Learning	Ensures data privacy while enabling efficient distributed learning.
[238–240]	Highlight advancements in network slicing for enabling customized and efficient resource allocation in IIoT.	Network Slicing	Network Slicing Technologies	Optimizes resource allocation and supports diverse IIoT use cases.
[241,242]	Examine quantum computing as a future enabler for complex problem-solving in IIoT scenarios.	Quantum Computing in IIoT	Quantum Algorithms and Systems	Addresses computational challenges in large-scale IIoT networks.

7. Conclusions

This survey provides a comprehensive analysis of the transformative impact of cloud computing in IIoT, offering insights into its architectures, service models, communication protocols, data management strategies, and application domains. The findings underscore the pivotal role cloud computing plays in enabling scalable, efficient, and intelligent IIoT systems.

Key architectural frameworks, such as centralized, distributed, and hybrid cloud models, were highlighted for their ability to address the computational and storage demands of IIoT environments. While centralized models facilitate large-scale data aggregation and analytics, distributed and hybrid architectures optimize latency and responsiveness, supporting real-time industrial processes. Service models like IaaS, PaaS, and SaaS provide

versatile solutions tailored to diverse operational needs, streamlining IIoT deployment and management.

The integration of advanced communication protocols, robust data management frameworks, and technologies such as Kubernetes and NFV ensures efficient data exchange and system reliability. Furthermore, applications like predictive maintenance, industrial automation, supply chain optimization, and energy management showcase the broad applicability of cloud computing in enhancing industrial operations. These advancements align with Industry 4.0 principles, fostering innovation through digital twins, robotic process automation, and autonomous process optimization.

Emerging challenges, such as security, scalability, and energy efficiency, emphasize the need for innovative solutions. The adoption of AI-driven analytics, federated learning, and sustainable computing practices presents a promising future for cloud-enabled IIoT systems. These findings highlight the critical importance of cloud computing in driving operational excellence and resilience, paving the way for a sustainable and digitally empowered industrial era.

Finally, future research could explore the implementation of adaptive cloud–edge hybrid models to address latency-sensitive IIoT applications and investigate blockchain integration for enhancing trust and transparency in data sharing. Moreover, advancing serverless computing for dynamic resource allocation and sustainable industrial operations presents a promising avenue for innovation.

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Abbreviations

The following abbreviations are used in this manuscript:

Acronym	Meaning
IIoT	Industrial Internet of Things
AI	Artificial intelligence
ML	Machine learning
MQTT	Message Queuing Telemetry Transport
MSE	Mean Squared Error
CoAP	Constrained Application Protocol
AMQP	Advanced Message Queuing Protocol
NFV	Network Function Virtualization
VNF	Virtualized network function
QoS	Quality of Service
GAN	Generative adversarial network
GDPR	General Data Protection Regulation
CCPA	California Consumer Privacy Act
IoT	Internet of Things
DQN	Deep Q-Network
PaaS	Platform as a Service
IaaS	Infrastructure as a Service
SaaS	Software as a Service
SDN	Software-defined networking
5G	Fifth-Generation Mobile Network
RPA	Robotic Process Automation

HVAC	Heating, ventilation, and air conditioning
CNN	Convolutional neural network
LSTM	Long short-term memory
RL	Reinforcement learning
DL	Deep learning
GPS	Global Positioning System
RFID	Radio-Frequency Identification
OPC-UA	Open Platform Communications Unified Architecture

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